



Optimal photosynthetic strategies

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Outline

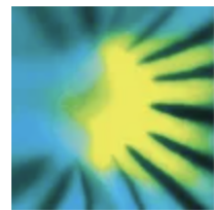
- Eco-evolutionary optimality theory
- Study 1: Global optimal photosynthetic strategies
- Study 2: Site-level optimal photosynthetic strategies
- Implications

Outline

- **Eco-evolutionary optimality theory**
- Study 1: Global optimal photosynthetic strategies
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Eco-evolutionary optimality theory

“EEO invokes the power of natural selection to eliminate uncompetitive trait combinations, and thereby shape predictable, general patterns in vegetation”



New Phytologist

Tansley review |  Free Access

Eco-evolutionary optimality as a means to improve vegetation and land-surface models

[Sandy P. Harrison](#) , [Wolfgang Cramer](#), [Oskar Franklin](#), [Iain Colin Prentice](#), [Han Wang](#), [Åke Brännström](#), [Hugo de Boer](#), [Ulf Dieckmann](#), [Jaideep Joshi](#), [Trevor F. Keenan](#), [Aliénor Lavergne](#) ... [See all authors](#) ▾

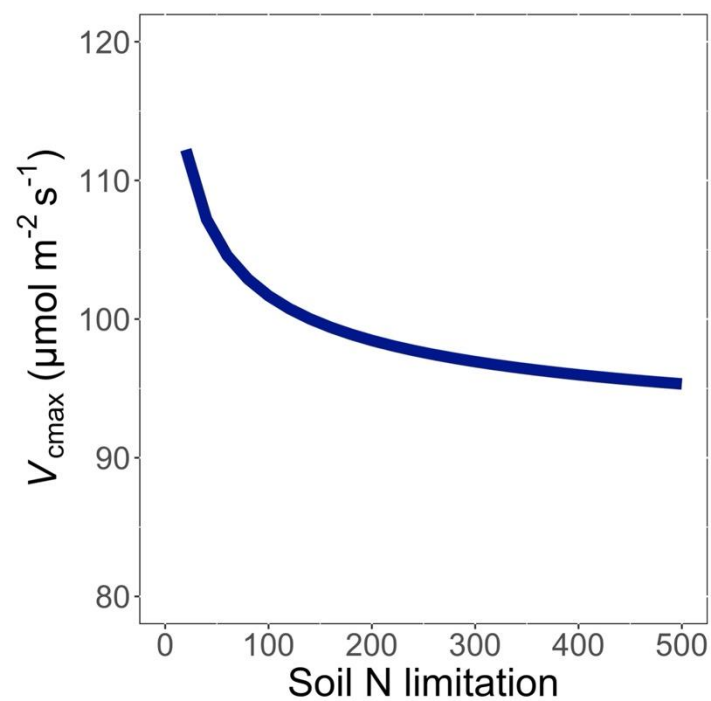
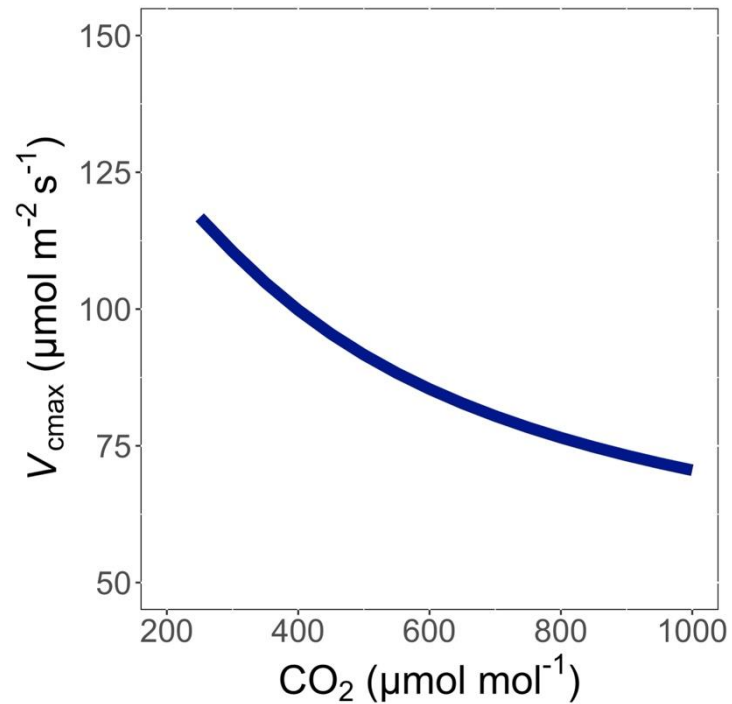
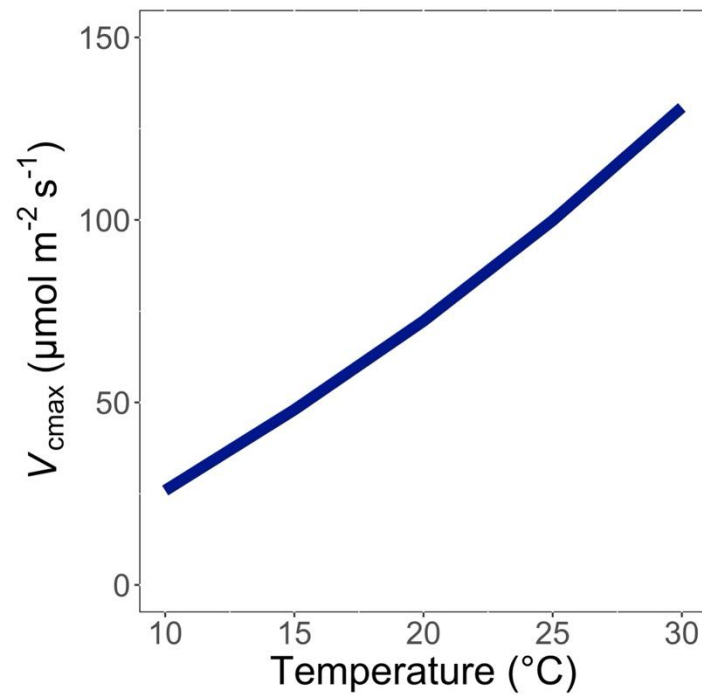
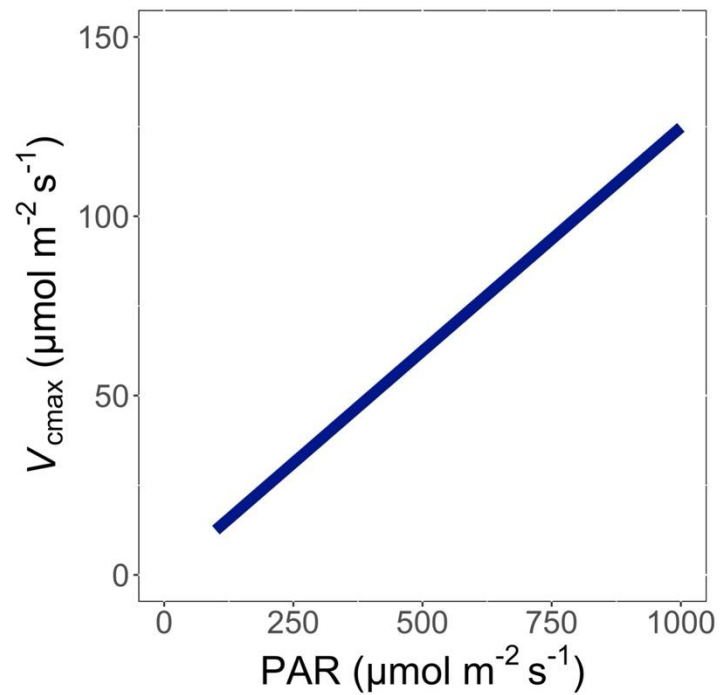
First published: 15 June 2021 | <https://doi.org/10.1111/nph.17558> |  VIEW METRICS

Photosynthesis optimization: least cost

Optimally, plants will maintain fastest rate of photosynthesis at the lowest summed resource cost (water and nutrient use)

Optimal trait example

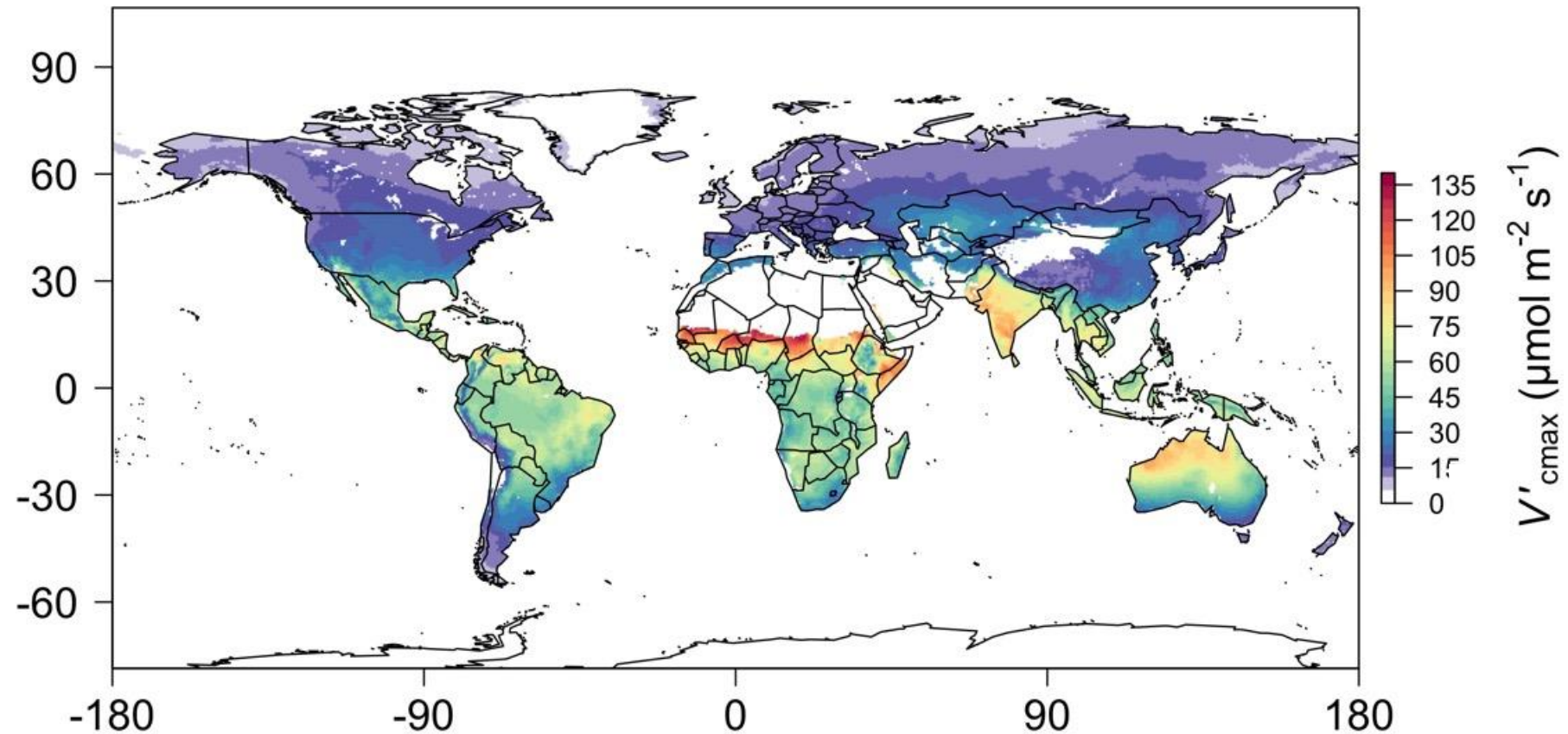
The maximum rate of RuBisCO
carboxylation (V_{cmax})



Trait-environment relationships can be predicted in ways that are:

1. Parameter-sparse
2. Mechanistically interpretable

We can predict optimal traits in different environments



We can predict optimal traits in different environments



Global Change Biology

PRIMARY RESEARCH ARTICLE | [Full Access](#)

Mechanisms underlying leaf photosynthetic acclimation to warming and elevated CO₂ as inferred from least-cost optimality theory

[Nicholas G. Smith](#) ✉ [Trevor F. Keenan](#)

First published: 11 June 2020 | <https://doi.org/10.1111/gcb.15212> | [VIEW METRICS](#)

JAMES | Journal of Advances in Modeling Earth Systems*

Research Article | [Open Access](#) | [CC](#) [i](#)

A Model of C₄ Photosynthetic Acclimation Based on Least-Cost Optimality Theory Suitable for Earth System Model Incorporation

[Helen G. Scott](#) ✉ [Nicholas G. Smith](#)

First published: 04 February 2022 | <https://doi.org/10.1029/2021MS002470> | [VIEW METRICS](#)

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RESEARCH ARTICLE | PLANT SCIENCES

Leaf economics fundamentals explained by optimality principles

[HAN WANG](#) ✉ [I. COLIN PRENTICE](#) [IAN J. WRIGHT](#) [DAVID I. WARTON](#) [SHENGCHAO QIAO](#) [XIANGTAO XU](#) [JIAN ZHOU](#) [KIHACHIRO KIKUZAWA](#) AND [NILS CHR. STENSETH](#) | [Authors Info & Affiliations](#)

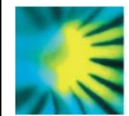
ECOLOGY LETTERS

Letter | [Open Access](#) | [CC](#) [i](#)

Global photosynthetic capacity is optimized to the environment

[Nicholas G. Smith](#) ✉ [Trevor F. Keenan](#) [I. Colin Prentice](#) [Han Wang](#) [Ian J. Wright](#) [Ülo Niinemets](#) [Kristine Y. Crous](#) [Tomas F. Domingues](#) [Rossella Guerrieri](#) [F. Yoko Ishida](#) [Jens Kattge](#) ... [See all authors](#)

First published: 04 January 2019 | <https://doi.org/10.1111/ele.13210> | [VIEW METRICS](#)



New Phytologist

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Reduced global plant respiration due to the acclimation of leaf dark respiration coupled with photosynthesis

[Yanghang Ren](#) [Han Wang](#) ✉ [Sandy P. Harrison](#) [I. Colin Prentice](#) [Owen K. Atkin](#) [Nicholas G. Smith](#) [Giulia Mengoli](#) [Artur Stefanski](#) [Peter B. Reich](#)

First published: 28 October 2023 | <https://doi.org/10.1111/nph.19355> | [VIEW METRICS](#)

How does the environment shape optimal photosynthesis in a multi-trait context?

How does the environment shape
optimal photosynthetic
strategies?

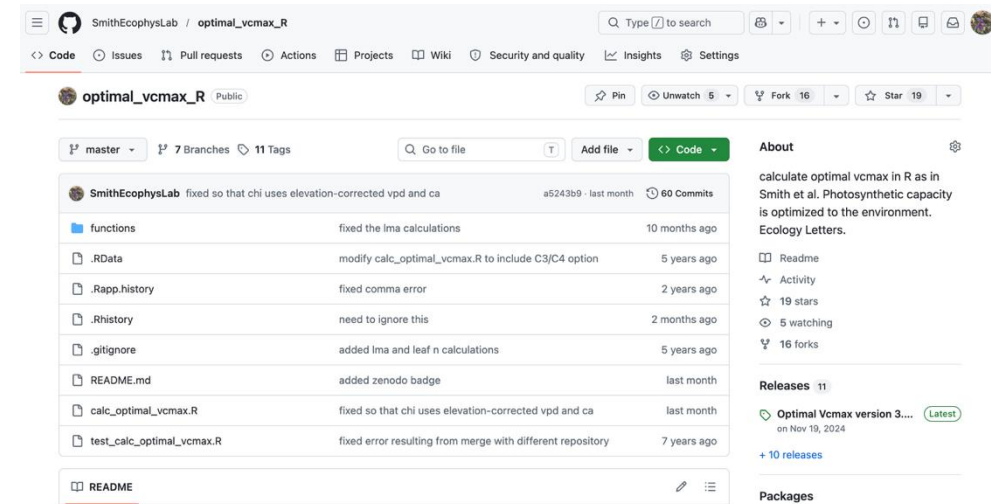
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Global simulation strategy

Global simulation strategy

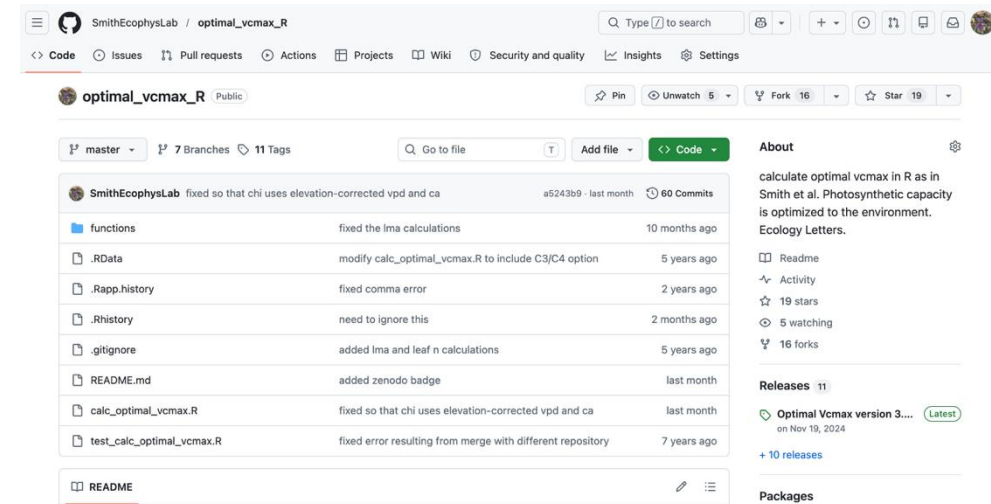
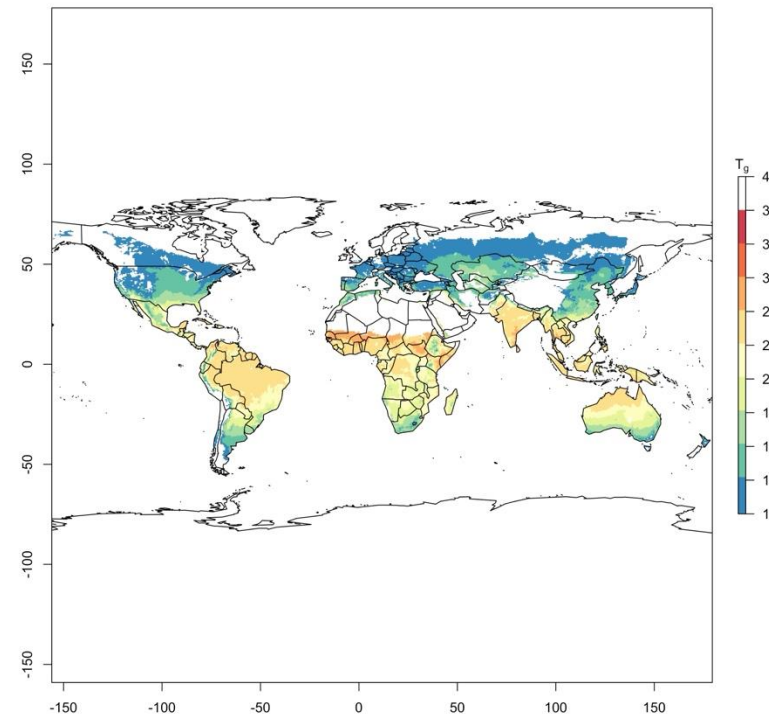
1. Developed optimal photosynthesis model



github.com/smithecophyslab/optimal_vcmax_r

Global simulation approach

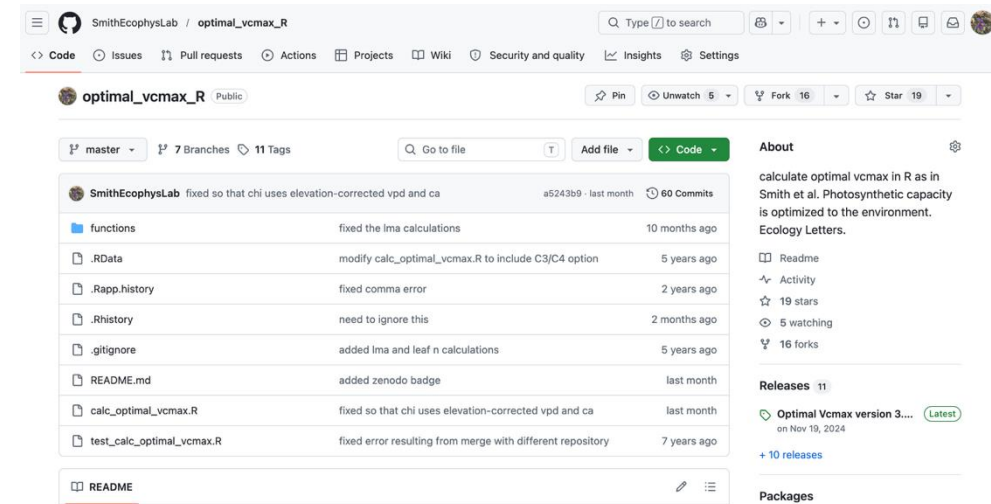
1. Developed optimal photosynthesis model
2. Used globally gridded climate to simulate traits

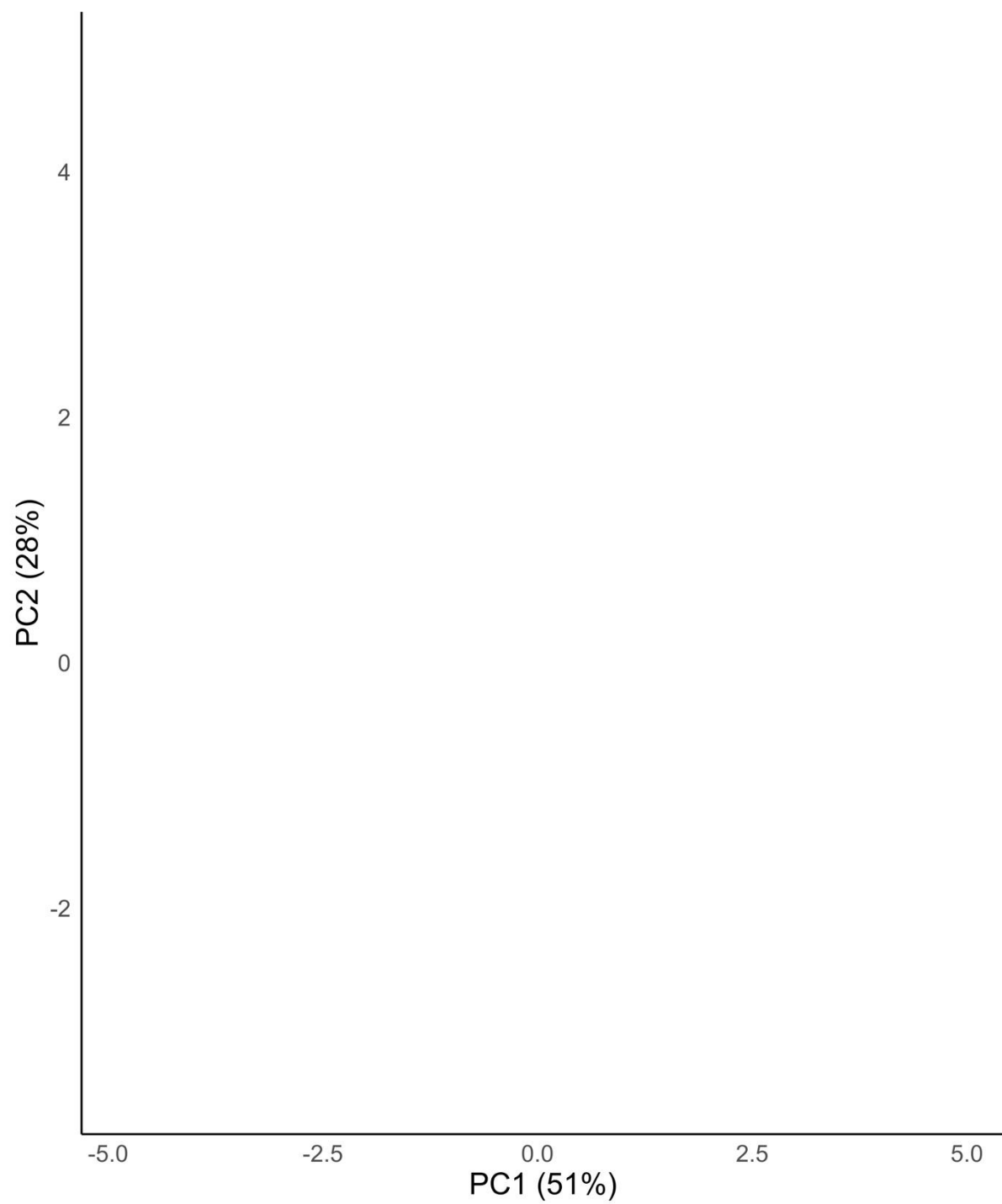


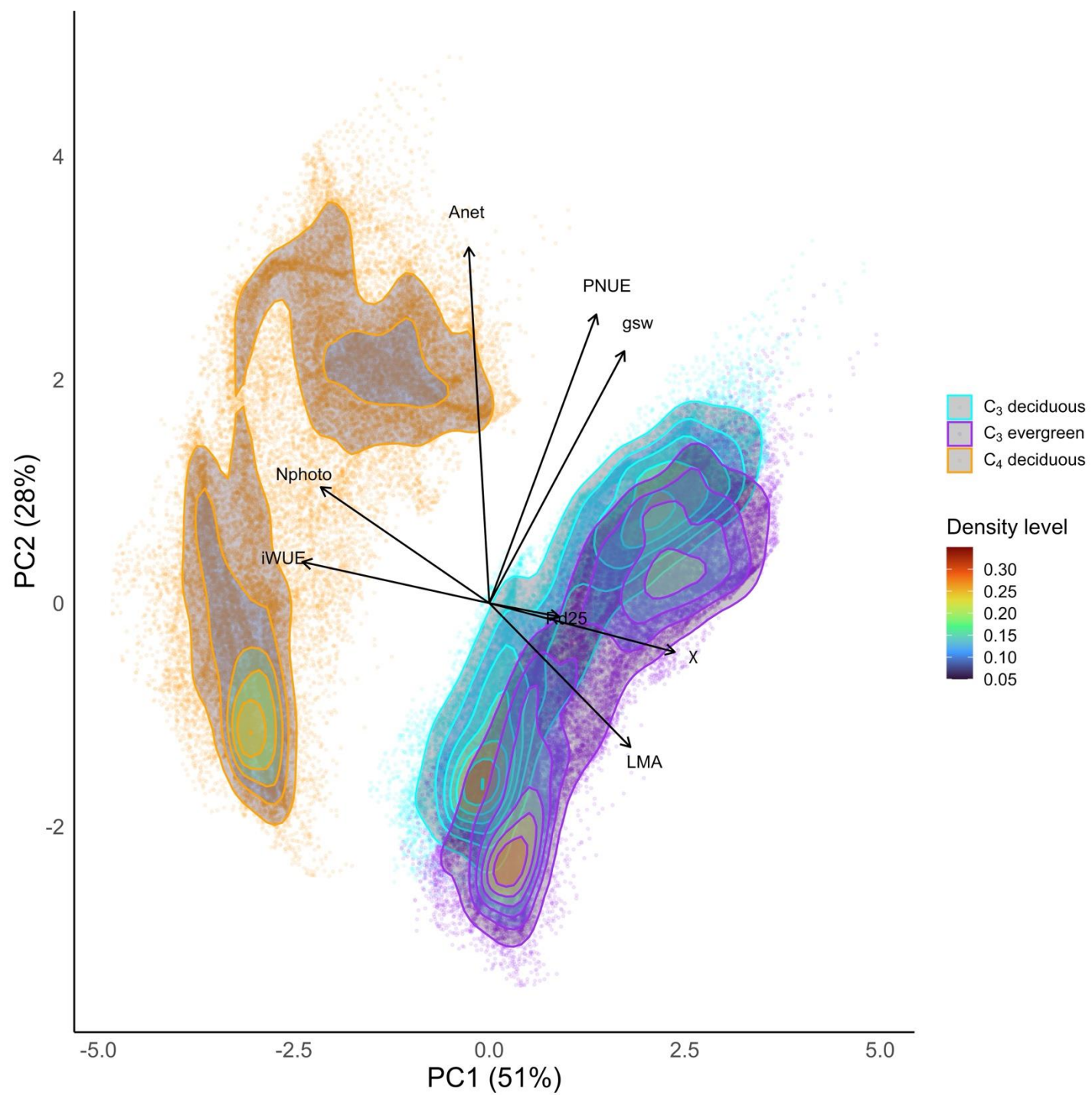
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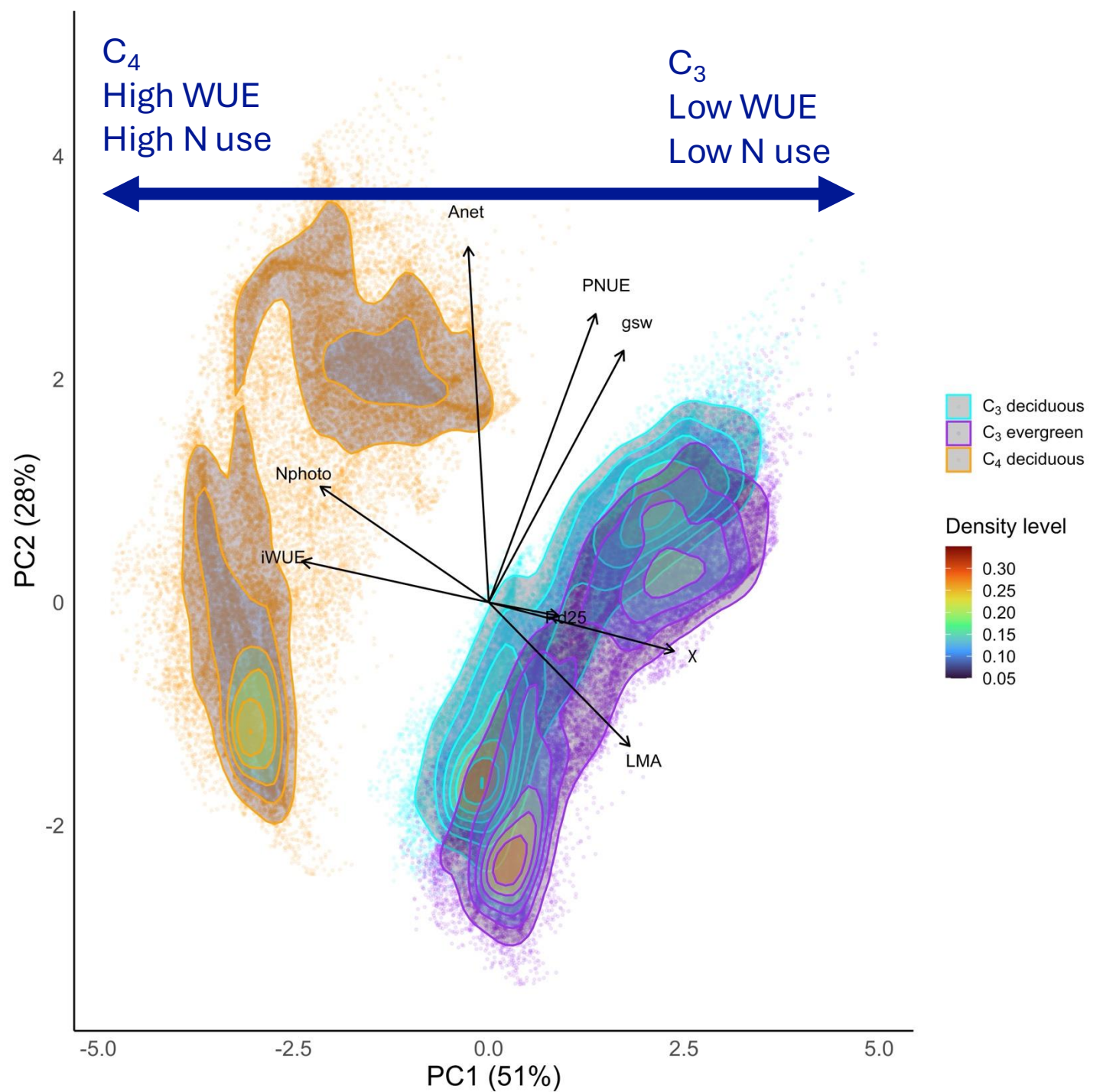
Global simulation approach

1. Developed optimal photosynthesis model
2. Used globally gridded climate to simulate traits
3. Used PCA to explore trait relationships and clustering
 - Carbon uptake (A_{net})
 - Carbon loss (R_d)
 - Water use for photosynthesis (g_{sw})
 - Nitrogen use for photosynthesis (N_{photo})
 - Leaf economics (LMA)
 - Least-cost theory (χ)
 - Photosynthetic efficiency ($iWUE$, $PNUE$)



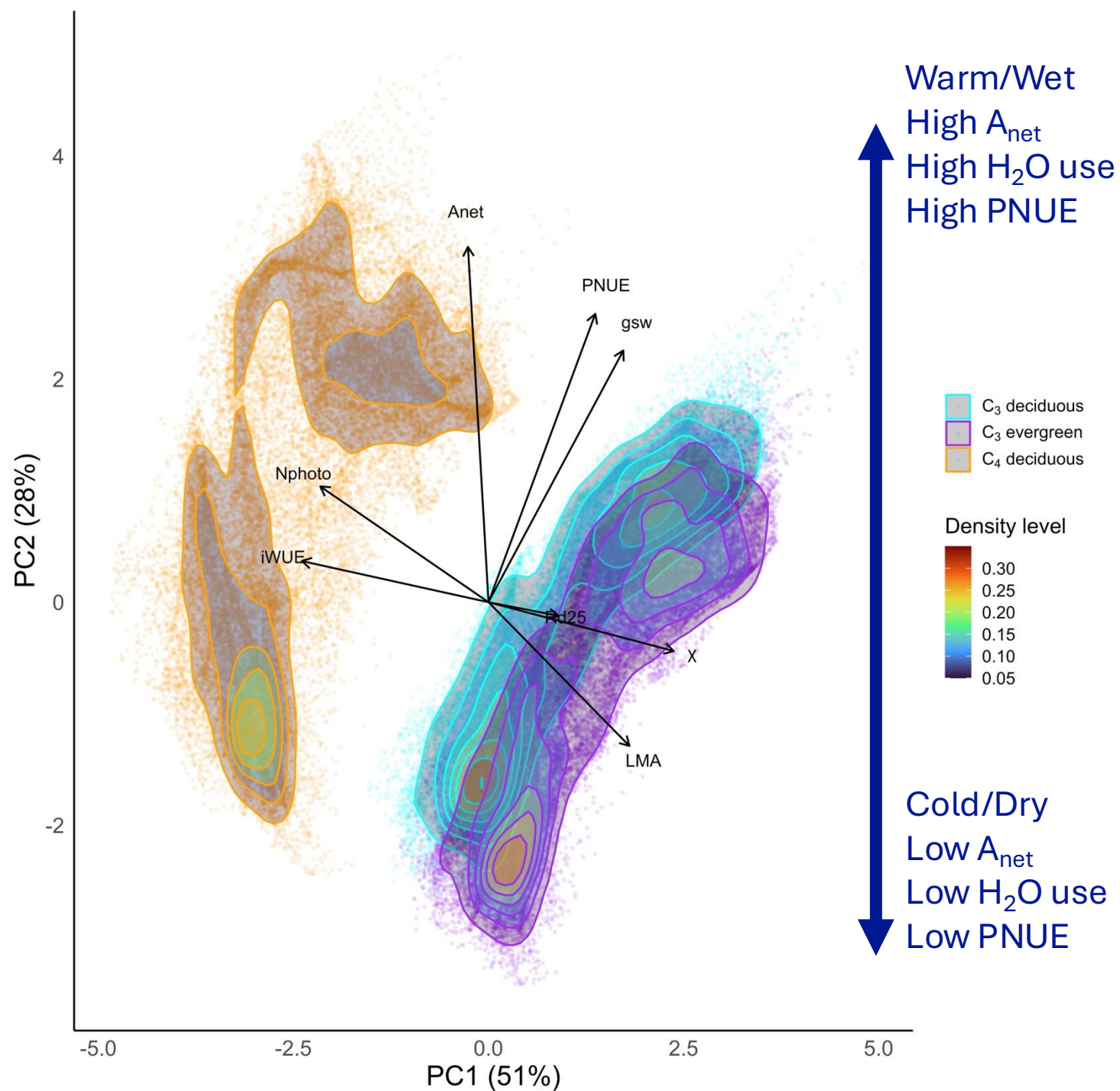






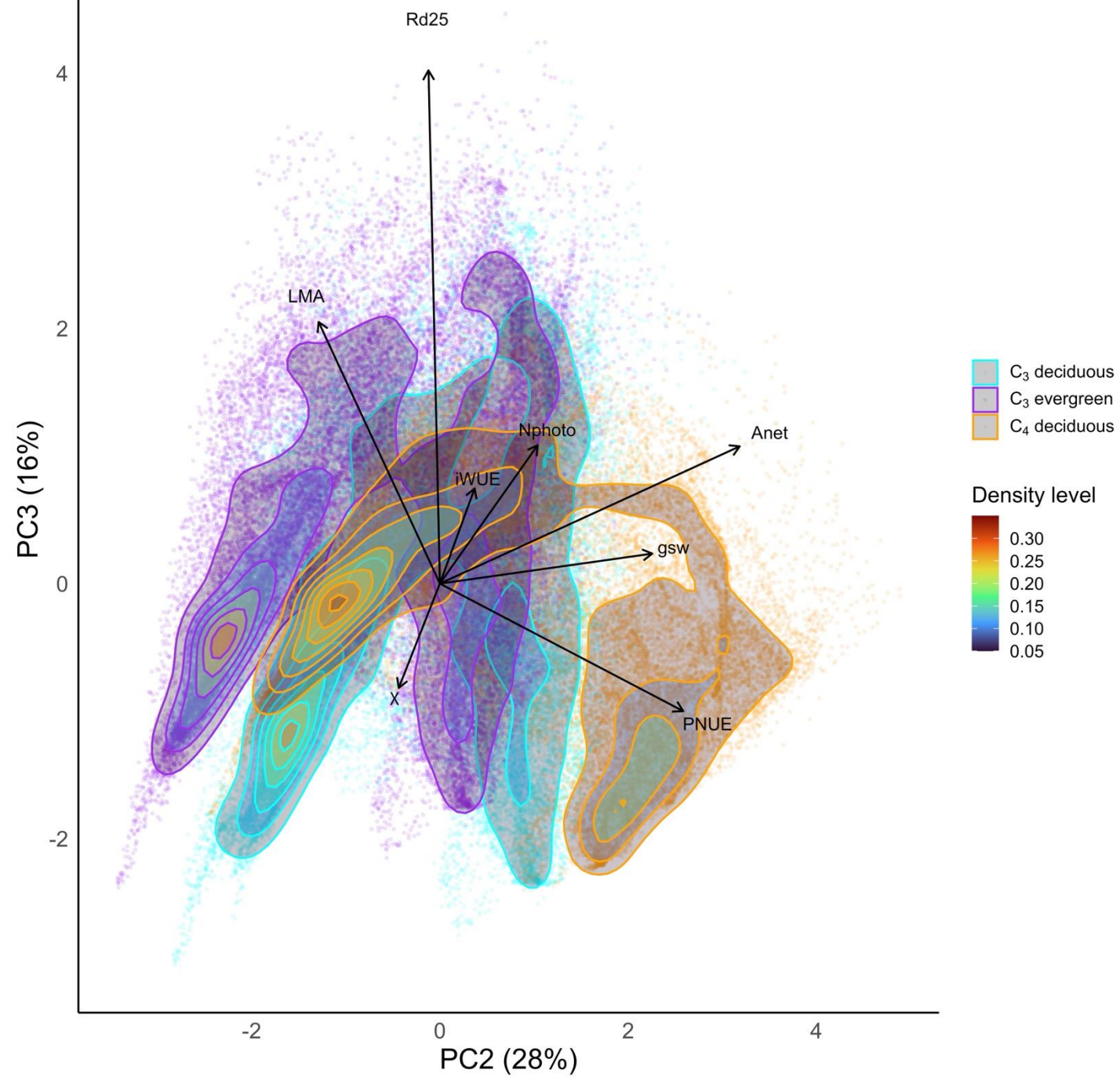
PC1 (51%)

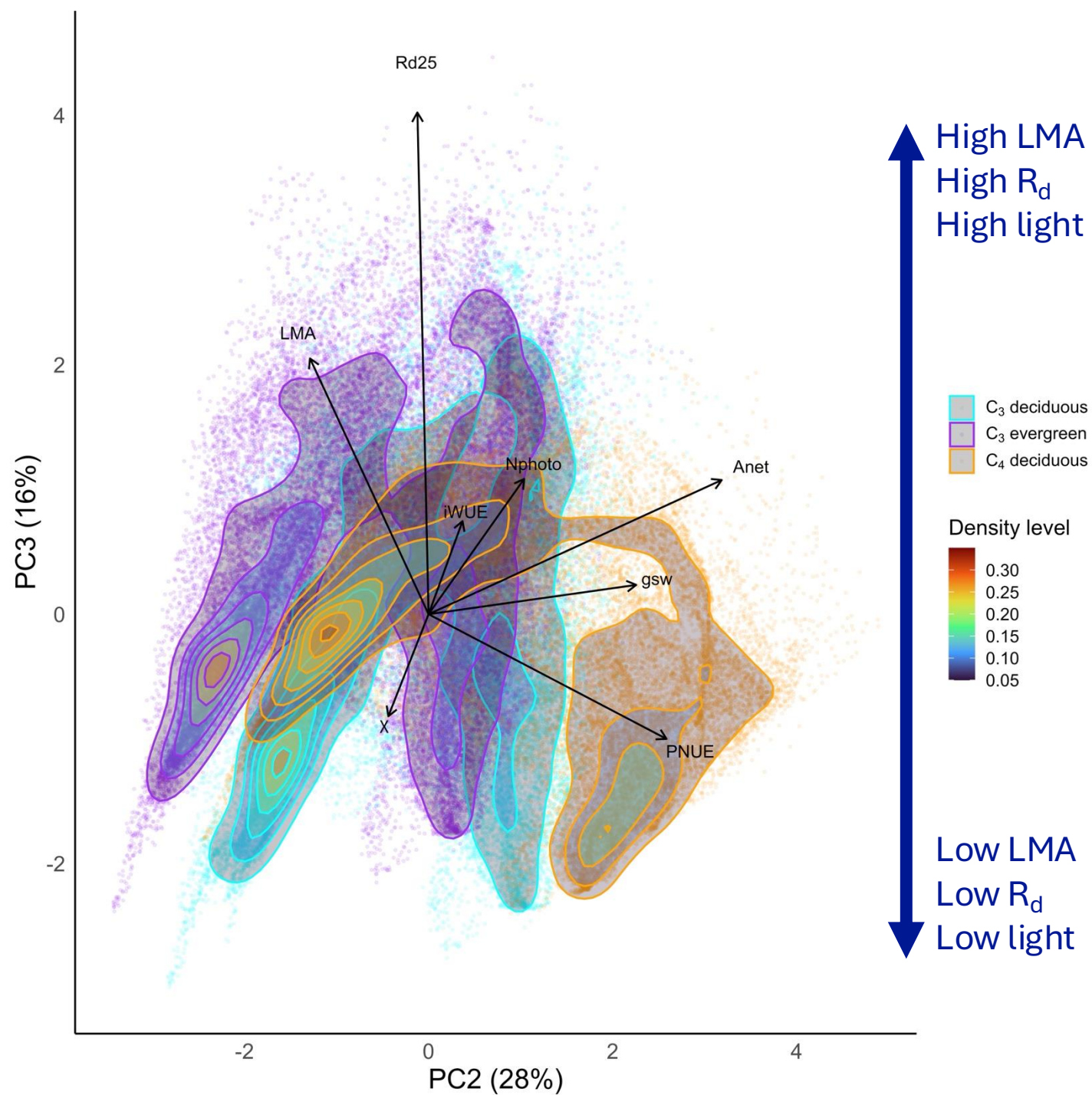
- N use-WUE
- C₃-C₄ divide



PC2 (28%)

- H_2O use-PNUE
- A_{net}
- Warm/wet-
Cold/dry divide





PC3 (16%)

- LMA, R_d (C costs)
- Light intensity

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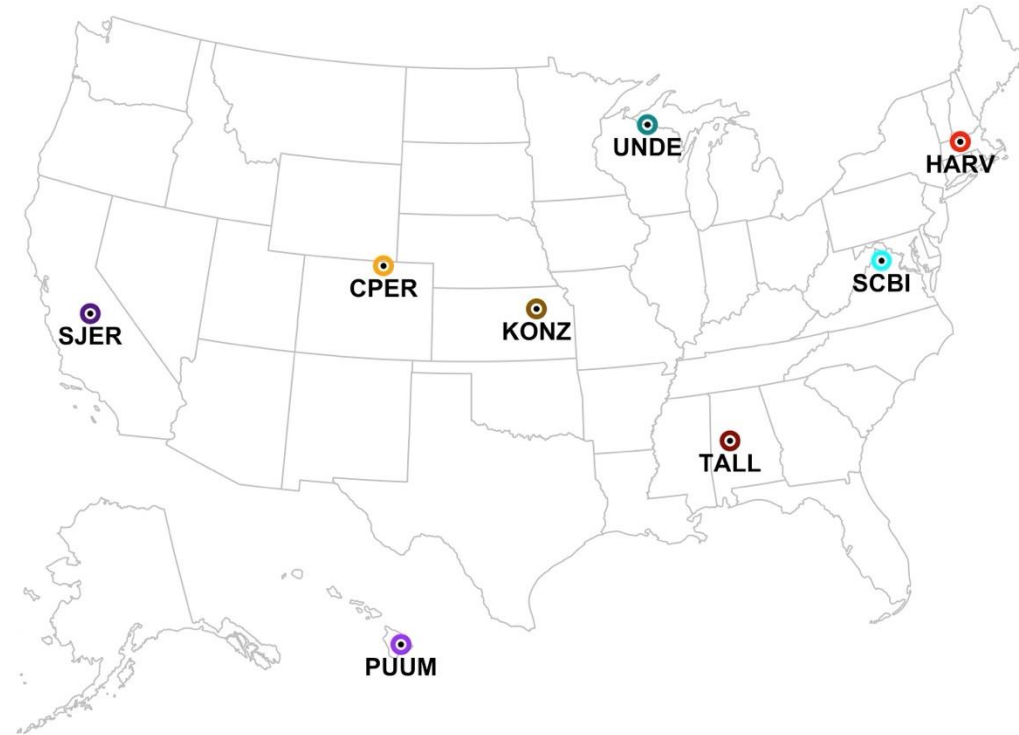
Site-level simulation approach

Site-level simulation approach

- Chose sites representing different dominant PFTs

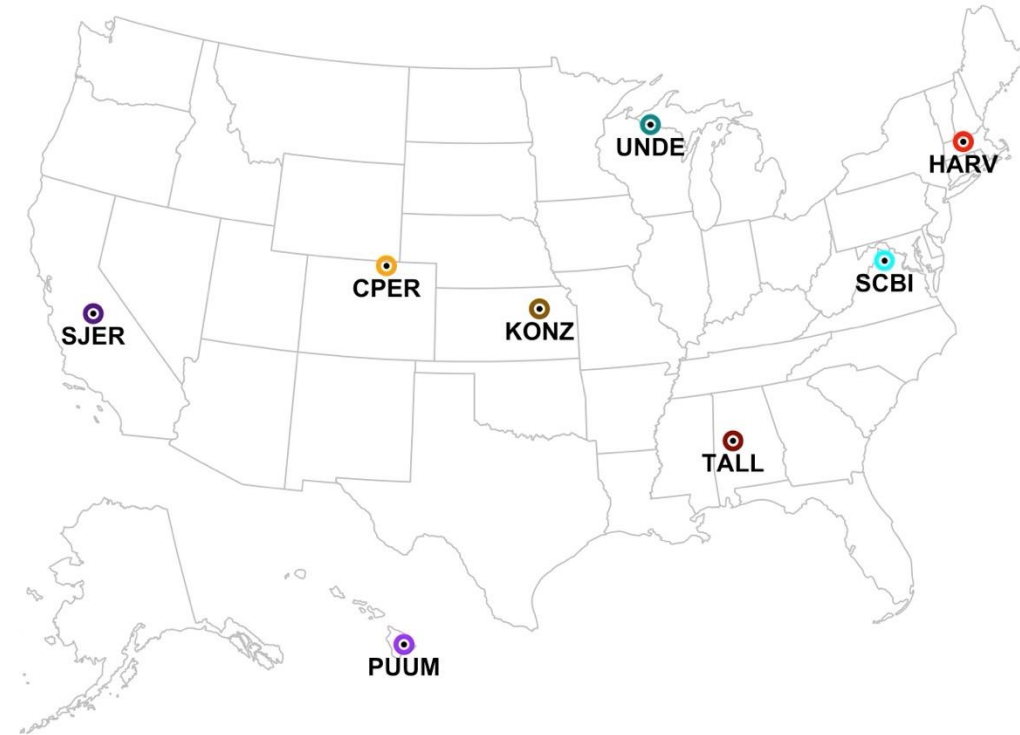
Site-level simulation approach

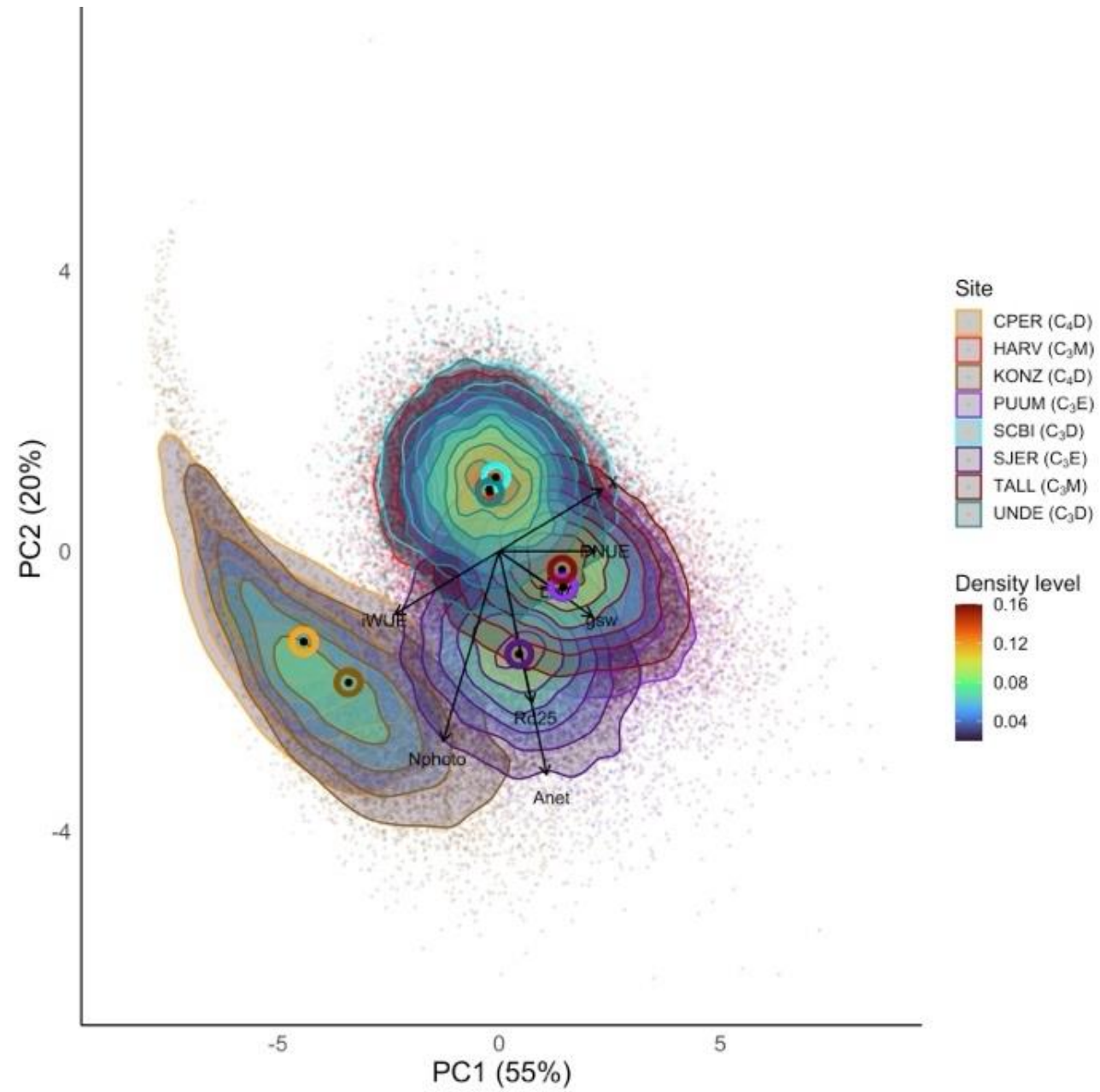
- Chose sites representing different dominant PFTs
 - NEON network
 1. Deciduous forests
 2. Evergreen forests
 3. Mixed forests
 4. C₄ grasslands

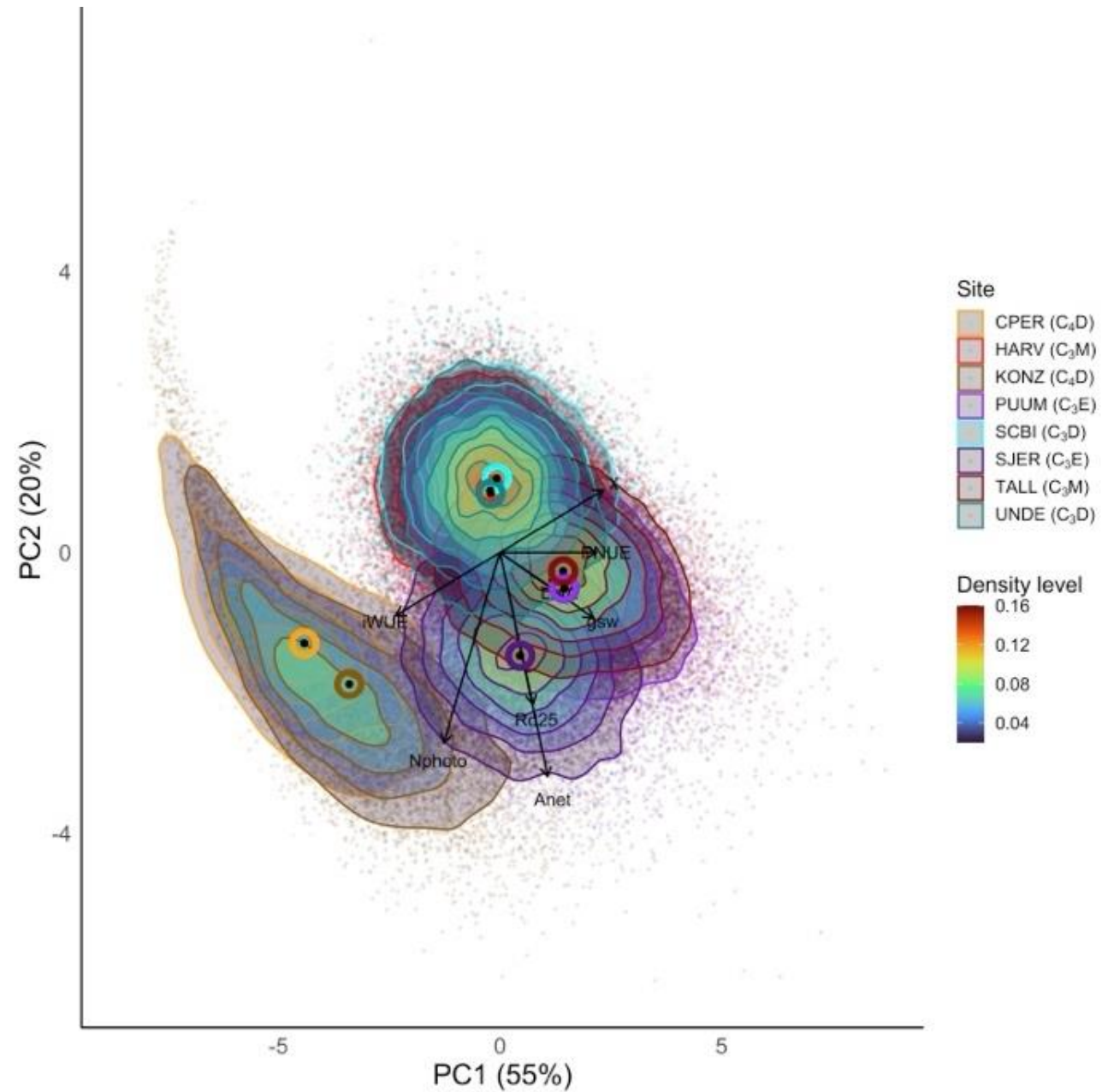


Site-level simulation approach

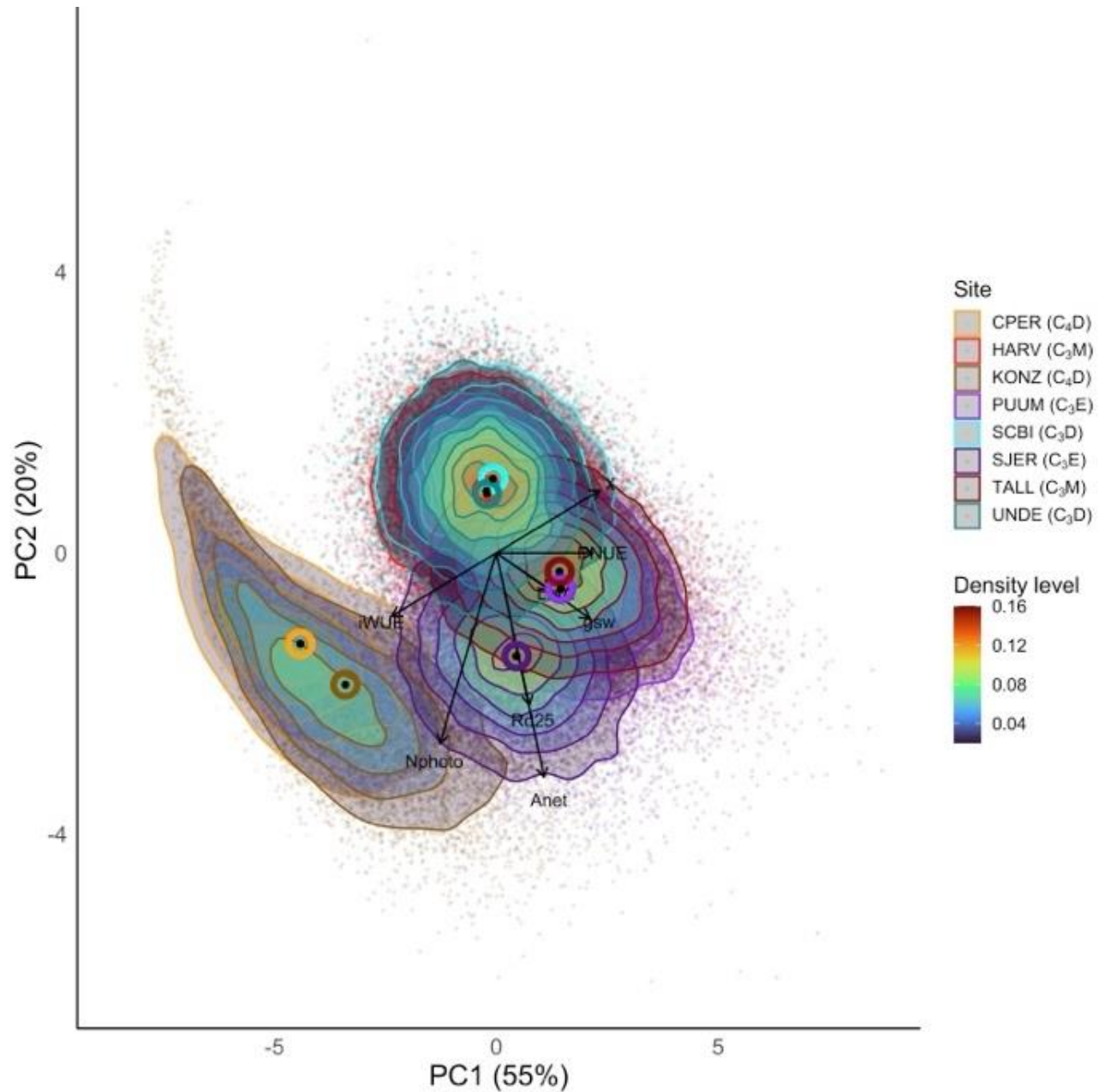
- Chose sites representing different dominant PFTs
 - NEON network
 1. Deciduous forests
 2. Evergreen forests
 3. Mixed forests
 4. C₄ grasslands
- Simulated optimal traits 10,000x using climate distribution
 - mean $\pm 0.1 \times \text{mean}$





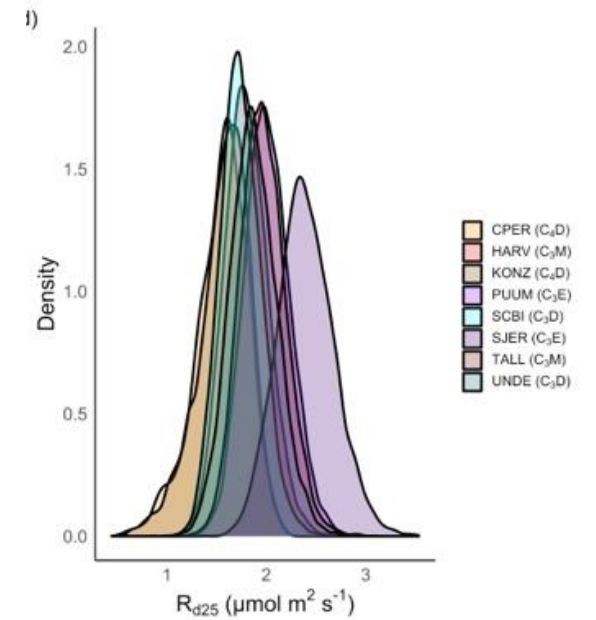
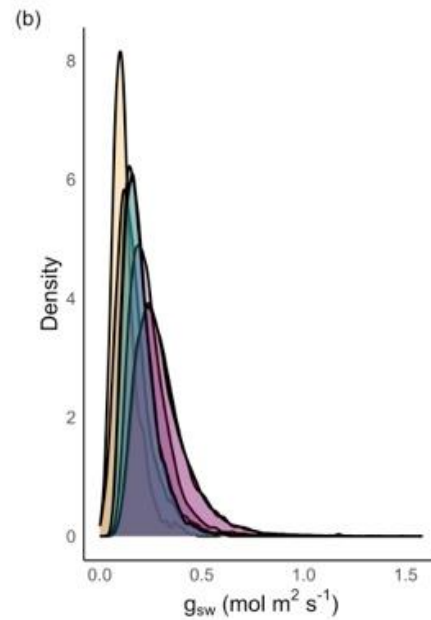
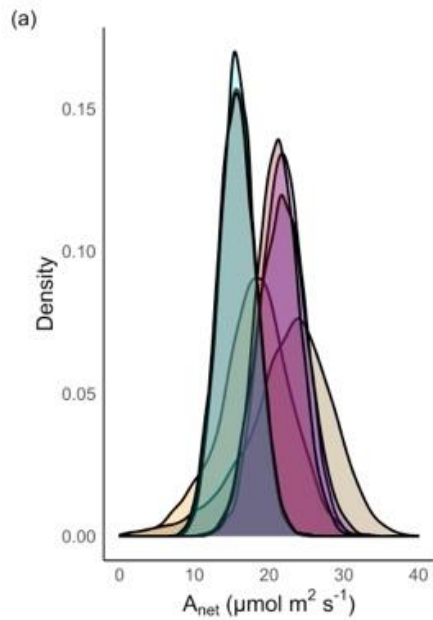


PCs defined similarly
to global simulation

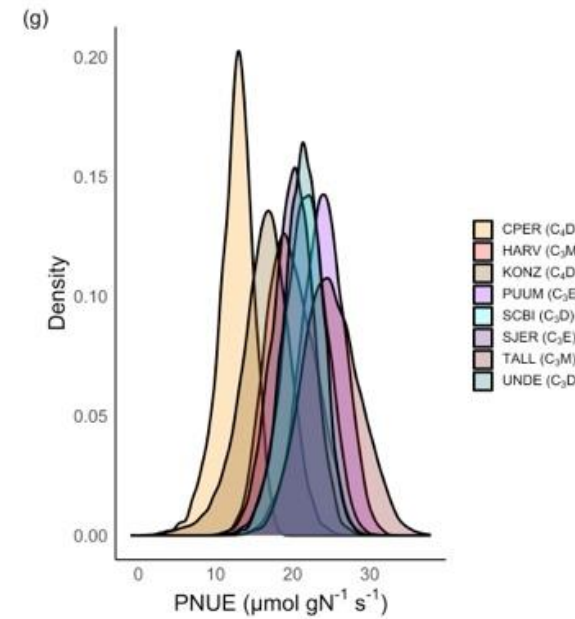


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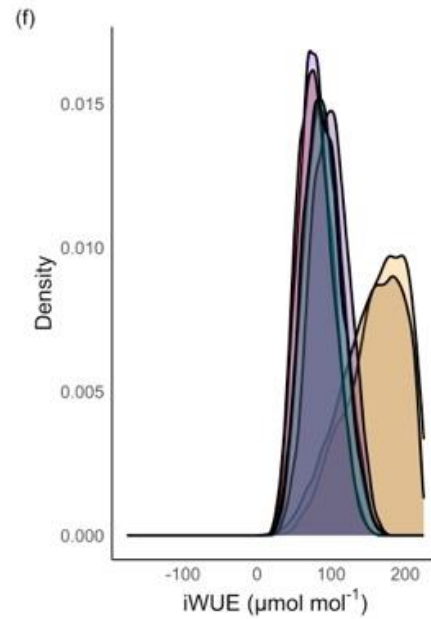
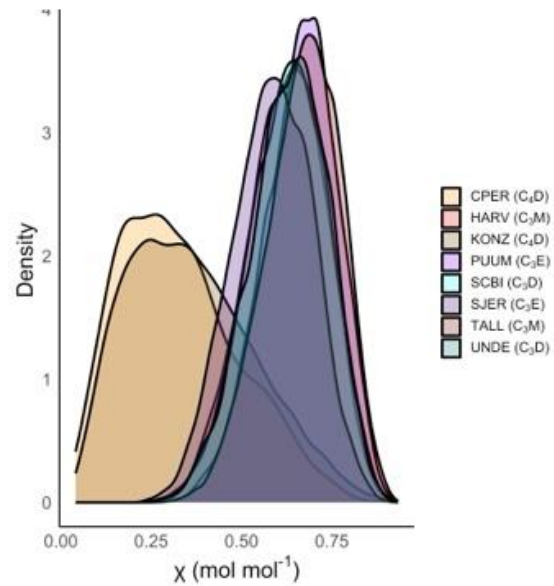
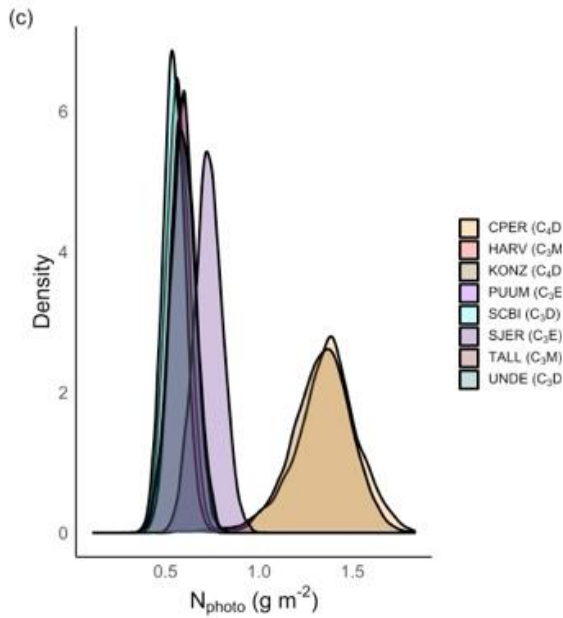
A lot of within-site
variability!



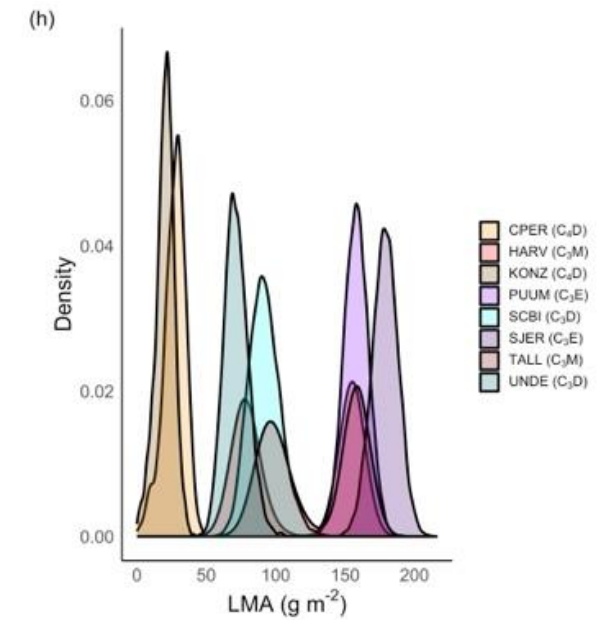
Within-site variability is larger than across-site for many traits



Some traits show
separation by
photosynthetic type



LMA shows separation by
PFT



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- Three defining axes (95%) of optimal photosynthetic strategies:

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 1. Nitrogen use-water use efficiency (51%; PFT-driven)

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 1. Nitrogen use-water use efficiency (51%; PFT-driven)
 2. Water use-nitrogen use efficiency (28%; climate-driven)

Implications

- We can simulate optimal photosynthetic strategies
- Three defining axes (95%) of optimal photosynthetic strategies:
 1. Nitrogen use-water use efficiency (51%; PFT-driven)
 2. Water use-nitrogen use efficiency (28%; climate-driven)
 3. Tissue costs of photosynthesis (16%; light-driven)

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 1. Nitrogen use-water use efficiency (51%; PFT-driven)
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- **Plant type** dictates optimal photosynthetic strategies via some (LMA, WUE, N use), but not all traits

Implications

- We can simulate optimal photosynthetic strategies
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 1. Nitrogen use-water use efficiency (51%; PFT-driven)
 2. Water use-nitrogen use efficiency (28%; climate-driven)
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- Plant type dictates optimal photosynthetic strategies via some (LMA, WUE, N use), but not all traits
- There are **many optimal strategies** within an environment

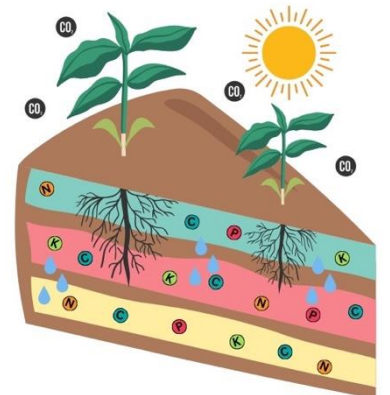
There are many ways plants can optimize photosynthesis and we can now quantitatively predict those strategies

Quick notice

- Lab is moving to the University of Nebraska - Lincoln
- Starting Fall 2026
- Looking for lab manager and graduate students
- Reach out if you are interested in collaborating



UNIVERSITY of NEBRASKA
LINCOLN



Presentation: github.com/SmithEcophysLab/esa_2026

Model code: github.com/SmithEcophysLab/optimal_vcm_max_r

Contact: nick.smith@ttu.edu; nsmith87@unl.edu



